Research and Engineering Portfolio

Student: **M. Yusuf Aykut**
Program: **B.Sc. Computer Science, Istanbul University**
Interests: **Medical robotics and physical AI**

ABOUT ME

I am a fourth-year Computer Science student at Istanbul University. My research started at my own university, where I used reinforcement learning to design optical devices with Assoc. Prof. Çağlar Çetinkaya. That work led to two published journal papers. I then worked at HaloScape on medical imaging models, first as an intern and then as a machine learning engineer. In summer 2026 I worked on proton therapy at Özyeğin University, locating where the beam stops from simulated sound signals, and at Koç University I measured how magnetic microswarms change shape under different magnetic fields.

Long term, I want to work on robotics and physical AI, mainly medical robotics. I am currently preparing a project on ultrasound Acoustic Phase Analysis (APA) with microswarms.

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Koç University, Medical Robotics Laboratory

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Istanbul University · two published papers

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01 · Profile

EXPERIENCE

Aug – Sep 2026
Robotics Research Intern
Koç University, Medical Robotics Laboratory
PI: Prof. Metin Sitti · Supervisor: Burhan Karahasan
Drove iron oxide microswarms with a 3-axis Helmholtz coil and built the video analysis that maps field settings to swarm shapes. Pages 3 and 4.

Jul – Aug 2026
Undergraduate Researcher
Özyeğin University, Summer Research Program
Supervisor: Prof. M. Burçin Ünlü
Simulation on patient CT data to locate where a proton beam stops from acoustic signals, and ranking of probe layouts. Page 5.

Jun 2025 – Mar 2026
Machine Learning Engineer
HaloScape · Jun 2025 – Mar 2026
ML Engineering Intern
HaloScape · Jun 2025 – Oct 2025
Industry, medical AI
Chest X-ray to CT generation, ECG foundation models, lumbar MRI disc grading, distributed PyTorch training on AWS. Page 6.

Apr 2024 – Aug 2025
Undergraduate Researcher
Istanbul University, Physics Department
Supervisor: Assoc. Prof. Çağlar Çetinkaya
Deep Q-learning over optical simulations for semi-transparent solar cell design. Two published journal papers. Page 7.

PUBLICATIONS

Çetinkaya, Ç., Çokduygulular, E., **Aykut, M. Y.**, Erkal, O., Aydoğmuş, F. & Kınacı, B. Artificial intelligence-empowered functional design of semi-transparent optoelectronic and photonic devices via deep Q-learning. *Scientific Reports* 15, 13508 (2025).

Çokduygulular, E., **Aykut, M. Y.** & Çetinkaya, Ç. Reinforcement learning approach to advance non-fullerene acceptor-based semi-transparent organic solar cell. *Advanced Theory and Simulations* 8(9), 2500334 (2025).

EDUCATION

B.Sc. Computer Science
Istanbul University, Faculty of Science
2023 – 2027 (expected) · 4th year

3.61 / 4.00 cumulative GPA

SELECTED COURSES

Signal and Image Processing	AA
Computer Vision	AA
Machine Learning	AA
Artificial Intelligence	AA
Linear Algebra	AA
Probability and Statistics	AA
Operating Systems	AA
Data Structures	AA

AA is the highest grade (4.0).

SKILLS

Programming · Python, C++, Java

Machine learning · PyTorch, TensorFlow, NumPy, Pandas, computer vision, reinforcement learning

Image processing · scikit-image, segmentation, moment-based shape analysis

Robotics and simulation · ROS, MuJoCo, Gazebo, TOPAS, k-Wave

Infrastructure · AWS, Docker, Flyte, Kubernetes, PostgreSQL

Languages · Turkish (native), English (fluent)

OTHER PROJECT

Intrinsic AI for Industry Challenge 2026. A fibre-connector insertion benchmark in the Gazebo robot simulator with a UR5e arm, and an imitation-learning policy trained on 420+ demonstrations.

ACTIVITIES

Founding member, Istanbul University Computer Science Club

Robotics member, ITU Perceptron Team

Organising team, Akademik Bilişim 2024 Conference

02 · Koç University, Medical Robotics Laboratory · Aug – Sep 2026

Morphological analysis and phase mapping of magnetic microswarms

PI Prof. Metin Sitti · Supervisor Burhan Karahasan

A microswarm is a large group of magnetic particles that moves as one body under an external magnetic field. Changing the field changes the shape the particles settle into: scattered clusters, ribbons or long chains. I wanted to classify these shapes with a number instead of by eye, and then map which field settings give which shape. For this I wrote an image processing pipeline that runs on every frame of the recordings.

SAMPLE	Iron oxide (Fe_3O_4) particles, 400 nm, 1 mg/mL, settled on the bottom of a petri dish
FIELD	A 3-axis Helmholtz coil around the dish. An oscillating field along x, $B_x = 80\ G \cdot \sin(2\pi ft)$, where f is the frequency in Hz. A constant offset B_y along y. No field along z ($B_z = 0$).
CONTROLS	The frequency f, and the amplitude ratio $\gamma = 80\ G / B_y$, which sets how widely the field direction sweeps
IMAGING	15 frames per second, 2592 × 1944 pixels, 8.2 µm per pixel, field of view 21.3 × 15.9 mm

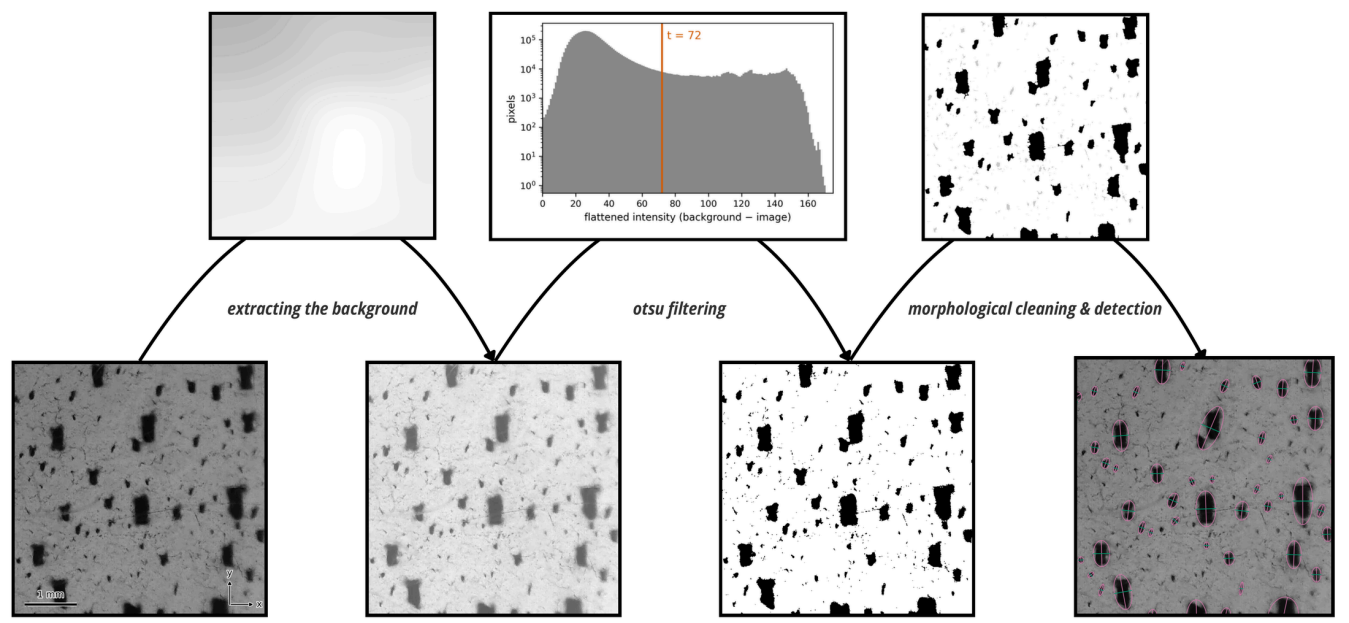


Figure 1. *Segmentation. The background is estimated and subtracted to remove uneven lighting. Otsu thresholding, a standard method that picks the brightness cut that best separates dark and light pixels, chose t = 72 here. Small fragments are then removed and each structure is detected.*

1 Objects

Touching dark pixels are grouped into one object, a set of pixel rows and columns (r_i, c_i). The mean row and column ($\bar{r}$, $\bar{c}$) give its centre.

2 Spread

μ_{20} and μ_{02} are the spread of the pixels along rows and columns, μ_{11} how the two vary together. They form the covariance matrix M.

3 Axes and angle

The eigenvalues $\lambda_1 \geq \lambda_2$ of M are the spread along the long and short axes. The tilt is $\theta = \frac{1}{2} \operatorname{atan2}(2\mu_{11}, \mu_{20} - \mu_{02})$.

4 Aspect ratio

Long axis $L_{\parallel} = 4\sqrt{\lambda_1}$, short axis $L_{\perp} = 4\sqrt{\lambda_2}$. The aspect ratio $\alpha = L_{\parallel} / L_{\perp}$ is 1 for a round object and large for a chain.

02 · Koç University · continued

Phase map

I ran 24 field conditions: four frequencies (10, 20, 40 and 60 Hz) and six values of γ from 2.00 to 13.33, set by stepping B_y from 40 to 6 G. Each condition was given one of three labels, using the aspect ratio and how the structures behaved in the video.

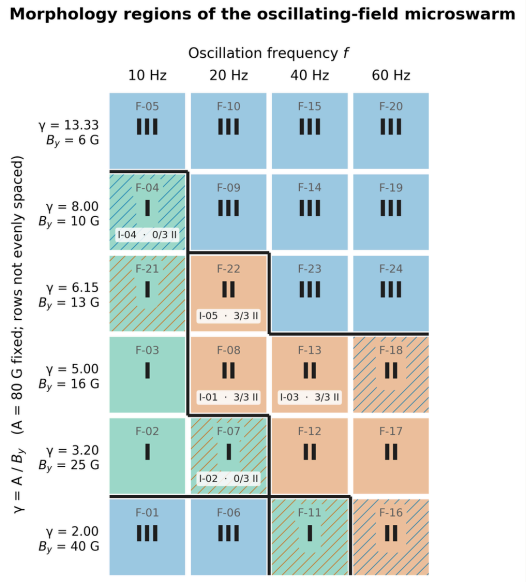


Figure 2. *Region of each field condition. Columns are frequencies, rows are values of γ (A is the 80 G oscillating amplitude). F-01 to F-24 are the condition numbers. Hatched cells lie close to a boundary. Tags such as "I-05 · 3/3 II" mark repeated conditions: three preparations, all three labelled Region II.*

THE THREE REGIONS

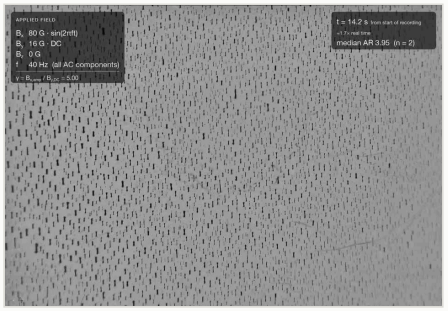
Region I	No ribbon yet. The particles form a disordered or chain-dominated pattern.
Region II	Ribbon-like swarm. One flexible body that quickly reshapes when the field changes.
Region III	Long chains that never close into a ribbon.

FINDINGS

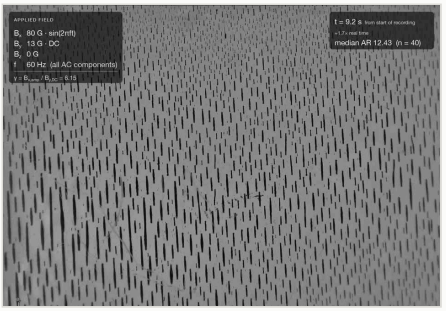
- The ribbon state (Region II) sits in a diagonal band that moves to lower frequency as γ rises. Above γ = 8 every frequency tested gave long chains.
- Five conditions were repeated with three newly prepared samples each. The measured values changed between samples, but no region label changed.
- The amount of particles in view varied about threefold between samples with no trend in aspect ratio, so the labels do not come from how much material was loaded.
- When the field is tilted against the dish floor, the ribbon rolls along the surface as one unit instead of falling apart, below 20 µm/s.



Region I · γ = 3.20 · f = 20 Hz



Region II · γ = 5.00 · f = 40 Hz



Region III · γ = 6.15 · f = 60 Hz

Figure 3. *Frames from the recordings, one per region. The overlay shows the applied field and the pipeline output: AR is the median aspect ratio in that frame and n the number of structures measured.*

Open point. Regions I and II can have similar aspect ratios. What separates them is how flexible the body is and how fast it reshapes, which a single-frame shape number does not capture. Making this boundary quantitative needs a measure taken over time.

Ongoing. I am continuing to prepare a follow-up project on ultrasound Acoustic Phase Analysis (APA) with these microswarms.

03 · Özyeğin University, Summer Research Program · Jul – Aug 2026

Locating the proton stopping point from sound

Supervisor Prof. M. Burçin Ünlü

A proton beam gives most of its dose where it stops, at a point called the Bragg peak. Where exactly it stops inside a patient is uncertain. When the protons stop they heat the tissue for a moment and send out a pressure wave, which ultrasound probes on the skin can record. I built this whole chain as a simulation on patient CT scans and used it to estimate the stopping point.

TOPAS, a Monte Carlo simulator that follows individual protons through the body, gives the dose D. The dose sets the starting pressure $p_0 = \Gamma \rho D$, where ρ is the tissue density and Γ (the Grüneisen parameter) says how much pressure a given amount of heat creates. k-Wave, an acoustic wave simulator, then carries this pressure through the tissue to the probes. From the probe signals I estimate the range $\hat{R}$, the depth where the beam stops.

CT scan + beam → TOPAS: dose D → $p_0 = \Gamma\rho D$ → k-Wave: sound → Probe signals → Range $\hat{R}$

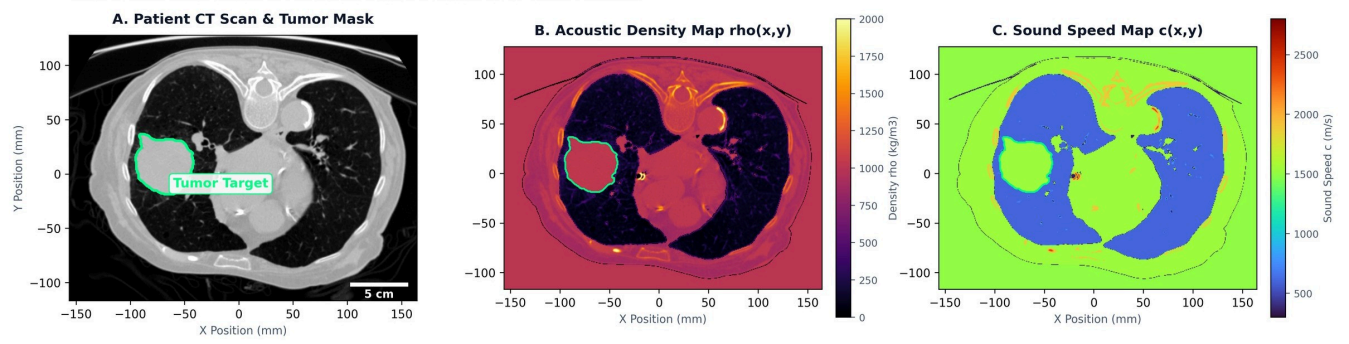


Figure 4. *CT values in Hounsfield units (HU) mapped voxel by voxel to density ρ and sound speed c. CT does not give these directly, so each tissue class gets a calibrated value, following Yu et al., Med. Phys. 46, 3649 (2019).*

I compared three ways to estimate the range: time of arrival, time reversal and waveform matching. I used waveform matching. It compares the full recorded waveform with simulated waveforms at known ranges, is the most precise of the three and gives an error bar.

PROBE PLACEMENT

Where the probes sit changes how well the stopping point can be pinned down. At planning time there is no measurement yet, so the true error cannot be minimised. For each candidate layout I moved the peak by 1 mm, measured how much each probe's waveform changed, and compared that change to the noise. This gives a predicted uncertainty for each layout before any measurement.

The ranking held. In all three patients the layout with the best predicted score also had the lowest range error in simulation. A poor but feasible layout raised the root-mean-square error (RMSE) by 1.8× to 9.3×, depending on the patient.

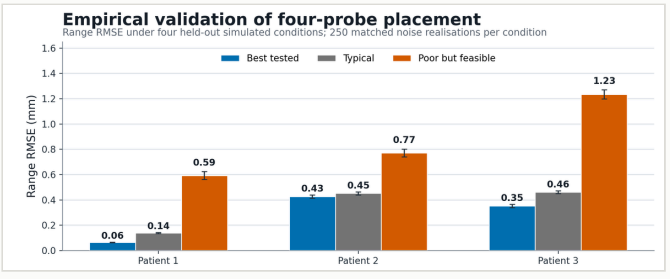


Figure 5. *Range RMSE in mm for the best tested, a typical and a poor but feasible probe layout, in three patients, with 250 noise draws per condition. Simulated error, not a clinical accuracy claim.*

04 · HaloScape · Jun 2025 – Mar 2026

Medical imaging models

Medical AI R&D

ML Engineering Intern

HaloScape

Jun 2025 – Oct 2025

Machine Learning Engineer

HaloScape

Jun 2025 – Mar 2026

WHAT I WORKED ON

- **Chest X-ray to CT.** Generative models that translate a single chest X-ray into CT, to estimate 3D anatomy from one planar scan.
- **ECG.** Fine-tuned ECG foundation models and built classifiers for cardiac anomaly detection with little labeled data.
- **Lumbar MRI.** A pipeline that finds each spinal disc in a lumbar MRI and grades how degenerated it is (Figure 6).
- **Training infrastructure.** Distributed PyTorch training on AWS, orchestrated with Flyte and Kubernetes.

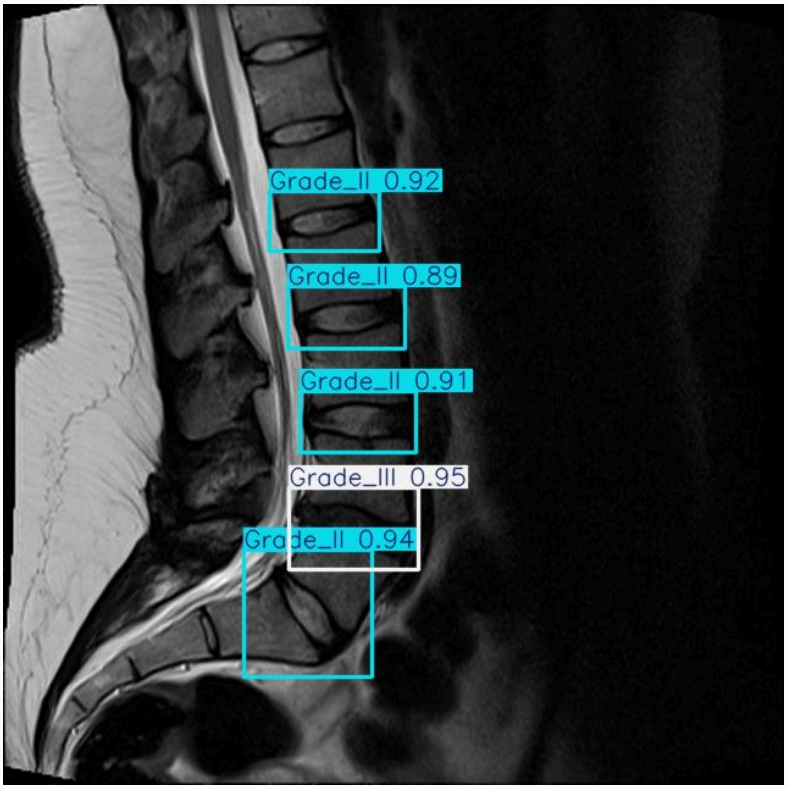


Figure 6. *Disc detection and degeneration grading on a sagittal lumbar MRI. Each box is one detected disc, labelled with its predicted degeneration grade and the model's confidence.*

05 · Istanbul University, Physics Department · Apr 2024 – Aug 2025

Reinforcement learning for optical device design

With Assoc. Prof. Çağlar Çetinkaya · two published journal papers

PUBLISHED PAPERS

1 Çetinkaya, Ç., Çokduygulular, E., **Aykut, M. Y.**, Erkal, O., Aydoğmuş, F. & Kınacı, B. Artificial intelligence-empowered functional design of semi-transparent optoelectronic and photonic devices via deep Q-learning. *Scientific Reports* 15, 13508 (2025). doi:10.1038/s41598-025-94586-x

2 Çokduygulular, E., **Aykut, M. Y.** & Çetinkaya, Ç. Reinforcement learning approach to advance non-fullerene acceptor-based semi-transparent organic solar cell. *Advanced Theory and Simulations* 8(9), 2500334 (2025). doi:10.1002/adts.202500334

My reinforcement learning work is also credited in Çokduygulular & Çetinkaya, Solar RRL 9(18), e202500455 (2025) and Solar Energy 307, 114339 (2026).

A semi-transparent solar cell has to let visible light through and still absorb enough light to make current. The two goals fight each other, and the result depends on the thickness of every layer in the stack. We measure them with the average visible transmittance (AVT), the share of visible light that passes through, and the photocurrent density J_{ph}.

I built and programmed a deep Q-learning agent that designs the cell. Deep Q-learning is a reinforcement learning method where a neural network learns which action is worth most in each state. Here the state holds the thickness and optical constants of the seven layers, and each action makes one layer 1 or 4 nm thicker or thinner. The environment is a transfer-matrix optical simulation, which computes how light reflects, passes and is absorbed in a layer stack. The reward pushes J_{ph} up and penalises designs below 25% AVT.

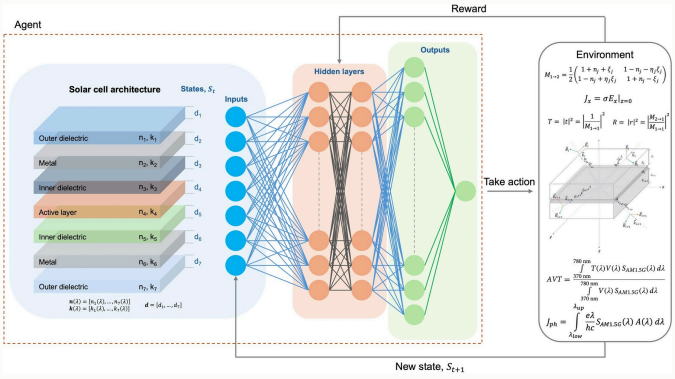


Figure 7. *Agent, network and environment. S_t is the state, d_1 to d_7 the layer thicknesses, n and k each layer's refractive index and extinction coefficient. Reproduced from paper 1, CC BY-NC-ND 4.0.*

BEST DESIGN FOUND

d_7	WO_3	top outer layer	33 nm
d_6	Ag	top thin metal	12 nm
d_5	MoO_3	hole transport	48 nm
d_4	PBDB-T:ITIC	organic active layer	93 nm
d_3	ZnO	electron transport	50 nm
d_2	Ag	bottom thin metal	5 nm
d_1	MgF_2	bottom outer layer	70 nm

31.58 J_{ph}, mA/cm²

25.20% AVT

Later I extended the workflow with a conditional diffusion model, a generative model that proposes candidate layer stacks for given target device properties.